

# Stepper and Servo Control Platform

Reusable embedded motion-control building blocks

EMBEDDED SYSTEMS

MOTION CONTROL

INSTRUMENTATION

## Project overview

A reusable bench platform for commanding, tuning, and comparing stepper and servo actuators. It is intended to support the chalk plotter, robotic arm, and other electromechanical prototypes.

## Engineering problem

Reliable motion requires more than issuing position commands: power integrity, timing, acceleration, calibration, feedback, thermal limits, and fault handling must work together.

### System architecture

- Command layer: position, velocity, acceleration, and coordinated-axis profiles.
- Driver layer: pulse/direction for steppers and calibrated pulse commands for servos.
- Safety layer: travel limits, enable control, current limits, timeouts, and stop behavior.
- Instrumentation: command/response logging and repeatable actuator test sequences.

### Engineering focus

- Compare open-loop stepper positioning with feedback-controlled servo behavior.
- Use acceleration ramps to reduce resonance, current spikes, and missed motion.
- Separate logic and motor power paths with deliberate grounding and decoupling.

## Verification plan

- Characterize position error, repeatability, maximum reliable speed, and thermal rise.
- Test limit-switch faults, interrupted commands, stalls, and restart behavior.
- Publish a clear pin map, calibration record, and interface contract for reuse.

### Status and evidence boundary

Summer 2026 controls platform brief supporting multiple physical prototypes.